

CAMBELL ISLAND, New Zealand

WMO: 939470

Lat: 52.550S

Lon: 169.150E

Elev: 19

StdP: 101.10

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-1.0	0.1	-5.3	2.4	1.0	-4.2	2.7	1.8	19.0	4.7	17.8	4.8	6.2	230	0.982

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	3.9	13.8	11.9	12.9	11.2	12.1	10.6	12.5	13.4	11.7	12.5	11.1	11.8	6.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.0	8.7	12.8	11.3	8.3	12.1	10.7	8.0	11.5	35.5	13.5	33.8	12.6	32.2	11.8	20.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	15.9	14.5		-2.8	16.4	1.4	1.8	-3.9	17.7	-4.7	18.7	-5.5	19.7	-6.5	20.9
				-3.5	14.2	1.3	1.7	-4.5	15.5	-5.3	16.5	-6.0	17.5	-7.0	18.8
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.0	9.2	9.2	8.7	7.7	6.5	5.3	5.1	5.4	5.7	6.0	6.8	8.6		
	DBStd	2.41	1.54	1.66	1.69	1.79	2.14	2.16	2.22	1.93	1.94	1.75	1.77	1.79		
	HDD10.0	1128	34	31	48	72	110	141	151	143	129	123	96	51		
	HDD18.3	4135	284	255	300	320	368	391	409	401	379	381	346	302		
	CDD10.0	34	9	9	6	2	1	0	0	0	0	0	1	7		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	8.1	8.2	8.2	8.2	8.5	8.2	8.4	8.0	7.5	7.9	8.4	8.1	7.6		
(15) Precipitation	PrecAvg	1342	121	108	120	116	130	108	105	106	112	111	99	108		
	PrecMax	1563	202	183	214	213	239	193	198	184	174	187	146	235		
	PrecMin	1072	43	59	39	54	58	57	37	41	31	59	39	31		
	PrecStd	129	38	32	36	33	40	31	32	31	33	29	27	45		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.8	15.6	14.6	12.7	11.7	10.2	9.9	10.0	10.2	11.2	12.1	14.8		
		MCWB	12.5	13.8	12.9	11.5	10.4	9.0	8.7	8.4	8.8	9.2	9.8	12.2		
	2%	DB	13.2	13.6	12.9	11.7	10.6	9.4	9.1	9.1	9.4	10.1	10.8	13.2		
		MCWB	11.2	12.1	11.6	10.8	9.7	8.5	8.2	7.9	8.3	8.5	8.8	11.1		
	5%	DB	12.3	12.5	11.9	10.9	10.0	8.9	8.6	8.6	8.9	9.4	10.1	12.2		
		MCWB	10.5	11.2	10.9	10.1	9.3	8.1	7.7	7.6	7.9	8.0	8.4	10.4		
	10%	DB	11.6	11.7	11.2	10.3	9.5	8.4	8.1	8.2	8.5	8.7	9.5	11.3		
		MCWB	10.0	10.5	10.1	9.5	8.8	7.6	7.3	7.3	7.5	7.4	8.1	9.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.1	14.0	13.1	12.0	10.8	9.5	9.1	8.9	9.2	9.9	10.6	12.8		
		MCDB	14.0	15.4	14.2	12.4	11.2	9.9	9.4	9.4	9.6	10.5	11.5	13.9		
	2%	WB	11.9	12.5	12.0	11.0	10.0	8.7	8.4	8.3	8.7	9.0	9.5	11.7		
		MCDB	12.8	13.3	12.7	11.4	10.4	9.2	8.8	8.8	9.1	9.7	10.2	12.8		
	5%	WB	11.1	11.6	11.2	10.3	9.5	8.2	7.9	7.9	8.2	8.3	8.9	10.8		
		MCDB	11.8	12.2	11.7	10.8	9.9	8.7	8.4	8.5	8.7	9.0	9.6	11.7		
	10%	WB	10.4	10.9	10.4	9.7	8.9	7.8	7.4	7.4	7.7	7.7	8.4	10.1		
		MCDB	11.3	11.5	11.0	10.2	9.4	8.3	8.0	8.1	8.3	8.5	9.2	11.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.0	3.9	3.5	3.3	3.3	3.3	3.1	3.3	3.5	4.0	4.0	4.2		
		MCDBR	5.0	4.8	4.6	3.6	3.5	3.2	3.1	3.3	3.4	4.4	4.5	5.2		
		MCWBR	3.9	4.2	4.2	3.8	3.7	3.5	3.3	3.3	3.4	3.9	3.6	3.8		
		MCDBR	4.4	4.4	4.3	3.7	3.6	3.2	3.0	3.2	3.2	3.9	3.8	4.7		
	5% WB	MCWBR	4.0	4.3	4.2	4.0	3.9	3.6	3.3	3.5	3.4	4.0	3.7	3.9		
(40) Clear-Sky Solar Irradiance	taub		0.341	0.339	0.335	0.328	0.316	0.310	0.312	0.325	0.352	0.343	0.337	0.340		
	taud		2.525	2.534	2.542	2.499	2.399	2.300	2.371	2.465	2.405	2.474	2.509	2.524		
	Ebn at Noon		963	929	869	767	641	560	632	754	830	913	961	974		
	Edh at Noon		103	95	82	68	55	50	55	69	92	100	104	105		
(44) All-Sky Solar Radiation	RadAvg		5.19	4.10	2.72	1.54	0.76	0.48	0.62	1.27	2.31	3.80	4.90	5.57		
	RadStd		0.33	0.28	0.21	0.08	0.06	0.03	0.05	0.10	0.16	0.17	0.21	0.29		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.25	N/A	+1.01	N/A	+0.53	+0.64	-90	-110	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page